

Raw scores to standard scores Ages 8 through 9				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
1	71	60	58	71
2	72	61	59	72
3	72	62	60	73
4	73	62	61	73
5	74	63	62	74
6	75	64	62	75
7	76	65	63	76
8	76	66	64	77
9	77	67	65	78
10	78	68	66	79
11	79	69	66	80
12	79	70	67	81
13	80	71	68	82
14	81	71	69	83
15	82	72	70	84
16	83	73	70	85
17	83	74	71	86
18	84	75	72	87
19	85	76	73	88
20	86	77	74	89
21	86	78	75	90
22	87	79	75	91
23	88	80	76	91
24	89	80	77	92
25	90	81	78	93
26	90	82	79	94
27	91	83	79	95
28	92	84	80	96
29	93	85	81	97
30	93	86	82	98

Raw scores to standard scores Ages 8 through 9				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
31	94	87	83	99
32	95	88	83	100
33	96	89	84	101
34	97	89	85	102
35	97	90	86	103
36	98	91	87	104
37	99	92	87	105
38	100	93	88	106
39	101	94	89	107
40	101	95	90	108
41	102	96	91	108
42	103	97	91	109
43	104	98	92	110
44	104	98	93	111
45	105	99	94	112
46	106	100	95	113
47	107	101	95	114
48	108	102	96	115
49	108	103	97	116
50	109	104	98	117
51	110	105	99	118
52	111	106	100	119
53	111	107	100	120
54	112	107	101	121
55	113	108	102	122
56	114	109	103	123
57	115	110	104	124
58	115	111	104	125
59	116	112	105	125
60	117	113	106	126

Raw scores to standard scores Ages 8 through 9				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
61	118	114	107	127
62	119	115	108	128
63	119	116	108	129
64	120	116	109	130
65	121	117	110	131
66	122	118	111	132
67	122	119	112	133
68	123	120	112	134
69	124	121	113	135
70	125	122	114	136
71	126	123	115	137
72	126	124	116	138
73	127	125	116	139
74	128	125	117	140
75	129	126	118	141
76	129	127	119	142
77	130	128	120	143
78	131	129	121	143
79	132	130	121	144
80	133	131	122	145
81	133	132	123	146
82	134	133	124	147
83	135	134	125	148
84	136	134	125	149
85	136	135	126	150
86	137	136	127	151
87	138	137	128	152
88	139	138	129	153
89	140	139	129	154
90	140	140	130	155

Raw scores to standard scores Ages 8 through 9				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
91	141	141	131	156
92	142	142	132	157
93	143	143	133	158
94	144	143	133	159
95	144	144	134	160
96	145	145	135	160
97	146	146	136	161
98	147	147	137	162
99	147	148	137	163
100	148	149	138	164

Learning Gain

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
-30	47	48	38	53
-29	48	49	39	54
-28	49	50	40	55
-27	49	51	41	56
-26	50	52	42	57
-25	51	53	43	58
-24	52	54	44	59
-23	53	55	45	60
-22	54	56	46	61
-21	55	57	47	61
-20	55	58	48	62
-19	56	59	49	63
-18	57	60	50	64
-17	58	61	51	65
-16	59	62	52	66
-15	60	63	53	67
-14	61	63	54	68
-13	61	64	55	69
-12	62	65	56	69
-11	63	66	57	70
-10	64	67	58	71
-9	65	68	59	72
-8	66	69	60	73
-7	67	70	61	74
-6	67	71	62	75
-5	68	72	63	76
-4	69	73	64	77
-3	70	74	65	77
-2	71	75	66	78
-1	72	76	67	79

Raw scores to standard scores Ages 8 through 9				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	73	77	68	80
1	73	78	69	81
2	74	79	70	82
3	75	80	71	83
4	76	81	72	84
5	77	81	73	85
6	78	82	74	86
7	79	83	75	86
8	79	84	76	87
9	80	85	77	88
10	81	86	78	89
11	82	87	79	90
12	83	88	80	91
13	84	89	81	92
14	85	90	82	93
15	85	91	83	94
16	86	92	84	94
17	87	93	85	95
18	88	94	86	96
19	89	95	87	97
20	90	96	88	98
21	91	97	89	99
22	91	98	90	100
23	92	99	91	101
24	93	100	92	102
25	94	100	93	102
26	95	101	94	103
27	96	102	95	104
28	97	103	96	105
29	97	104	97	106

Learning Gain

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	98	105	98	107
31	99	106	99	108
32	100	107	100	109
33	101	108	101	110
34	102	109	102	110
35	103	110	103	111
36	103	111	104	112
37	104	112	105	113
38	105	113	106	114
39	106	114	107	115
40	107	115	108	116
41	108	116	109	117
42	109	117	110	118
43	109	118	111	119
44	110	119	112	119
45	111	119	113	120
46	112	120	114	121
47	113	121	115	122
48	114	122	116	123
49	115	123	117	124
50	115	124	118	125
51	116	125	119	126
52	117	126	120	127
53	118	127	121	127
54	119	128	122	128
55	120	129	122	129
56	121	130	123	130
57	121	131	124	131
58	122	132	125	132
59	123	133	126	133

Raw scores to standard scores Ages 8 through 9				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	124	134	127	134
61	125	135	128	135
62	126	136	129	135
63	127	137	130	136
64	127	137	131	137
65	128	138	132	138
66	129	139	133	139
67	130	140	134	140
68	131	141	135	141
69	132	142	136	142
70	133	143	137	143
71	133	144	138	143
72	134	145	139	144
73	135	146	140	145
74	136	147	141	146
75	137	148	142	147

Delayed recall

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	74	59	64	69
1	74	60	65	70
2	75	61	65	71
3	75	62	66	71
4	76	62	67	72
5	77	63	67	73
6	77	64	68	74
7	78	65	68	75
8	78	65	69	76
9	79	66	69	76
10	80	67	70	77
11	80	68	71	78
12	81	68	71	79
13	81	69	72	80
14	82	70	72	81
15	83	71	73	81
16	83	71	74	82
17	84	72	74	83
18	84	73	75	84
19	85	74	75	85
20	86	74	76	86
21	86	75	76	87
22	87	76	77	87
23	87	77	78	88
24	88	77	78	89
25	89	78	79	90
26	89	79	79	91
27	90	80	80	92
28	90	80	80	92
29	91	81	81	93

Raw scores to standard scores Ages 8 through 9				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	92	82	82	94
31	92	83	82	95
32	93	84	83	96
33	93	84	83	97
34	94	85	84	97
35	94	86	85	98
36	95	87	85	99
37	96	87	86	100
38	96	88	86	101
39	97	89	87	102
40	97	90	87	103
41	98	90	88	103
42	99	91	89	104
43	99	92	89	105
44	100	93	90	106
45	100	93	90	107
46	101	94	91	108
47	102	95	92	108
48	102	96	92	109
49	103	96	93	110
50	103	97	93	111
51	104	98	94	112
52	105	99	94	113
53	105	99	95	113
54	106	100	96	114
55	106	101	96	115
56	107	102	97	116
57	108	102	97	117
58	108	103	98	118
59	109	104	99	119

Delayed recall

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	109	105	99	119
61	110	105	100	120
62	111	106	100	121
63	111	107	101	122
64	112	108	101	123
65	112	109	102	124
66	113	109	103	124
67	114	110	103	125
68	114	111	104	126
69	115	112	104	127
70	115	112	105	128
71	116	113	106	129
72	117	114	106	129
73	117	115	107	130
74	118	115	107	131
75	118	116	108	132
76	119	117	108	133
77	119	118	109	134
78	120	118	110	135
79	121	119	110	135
80	121	120	111	136
81	122	121	111	137
82	122	121	112	138
83	123	122	113	139
84	124	123	113	140
85	124	124	114	140
86	125	124	114	141
87	125	125	115	142
88	126	126	115	143
89	127	127	116	144

Raw scores to standard scores Ages 8 through 9				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
90	127	127	117	145
91	128	128	117	146
92	128	129	118	146
93	129	130	118	147
94	130	131	119	148
95	130	131	120	149
96	131	132	120	150
97	131	133	121	151
98	132	134	121	151
99	133	134	122	152
100	133	135	122	153

Forgetting

Ages 8 through 9

Raw scores to standard scores Ages 8 through 9				
Forgetting Raw	Verbal	Non- verbal	Semantic	Non- semantic
-50	1	16	25	33
-49	1	17	26	35
-48	1	19	28	36
-47	3	21	30	38
-46	6	23	32	39
-45	8	25	33	41
-44	10	26	35	42
-43	13	28	37	44
-42	15	30	38	45
-41	18	32	40	47
-40	20	34	42	48
-39	23	36	44	49
-38	25	37	45	51
-37	28	39	47	52
-36	30	41	49	54
-35	33	43	51	55
-34	35	45	52	57
-33	38	46	54	58
-32	40	48	56	60
-31	42	50	57	61
-30	45	52	59	63
-29	47	54	61	64
-28	50	55	63	66
-27	52	57	64	67
-26	55	59	66	69
-25	57	61	68	70
-24	60	63	69	71
-23	62	64	71	73
-22	65	66	73	74
-21	67	68	75	76

Raw scores to standard scores Ages 8 through 9				
Forgetting Raw	Verbal	Non- verbal	Semantic	Non- semantic
-20	70	70	76	77
-19	72	72	78	79
-18	74	73	80	80
-17	77	75	82	82
-16	79	77	83	83
-15	82	79	85	85
-14	84	81	87	86
-13	87	82	88	88
-12	89	84	90	89
-11	92	86	92	90
-10	94	88	94	92
-9	97	90	95	93
-8	99	91	97	95
-7	102	93	99	96
-6	104	95	100	98
-5	106	97	102	99
-4	109	99	104	101
-3	111	100	106	102
-2	114	102	107	104
-1	116	104	109	105
0	119	106	111	107
1	121	108	112	108
2	124	109	114	110
3	126	111	116	111
4	129	113	118	112
5	131	115	119	114
6	134	117	121	115
7	136	118	123	117
8	138	120	125	118
9	141	122	126	120

Raw scores to standard scores Ages 8 through 9				
Forgetting Raw	Verbal	Non- verbal	Semantic	Non- semantic
10	143	124	128	121
11	146	126	130	123
12	148	128	131	124
13	151	129	133	126
14	153	131	135	127
15	156	133	137	129
16	158	135	138	130
17	161	137	140	131
18	163	138	142	133
19	166	140	143	134

Recognition
Ages 8 through 9

Raw scores to standard scores Ages 8 through 9				
Recognition Raw	Verbal	Non-verbal	Semantic	Non-semantic
-30	54	39	29	57
-29	54	40	30	57
-28	54	41	30	57
-27	55	41	31	58
-26	55	42	32	58
-25	56	42	32	59
-24	56	43	33	59
-23	57	44	34	60
-22	57	44	34	60
-21	58	45	35	61
-20	58	45	36	61
-19	59	46	36	62
-18	59	47	37	62
-17	60	47	38	63
-16	60	48	38	63
-15	61	48	39	63
-14	61	49	40	64
-13	62	50	40	64
-12	62	50	41	65
-11	62	51	42	65
-10	63	51	42	66
-9	63	52	43	66
-8	64	52	44	67
-7	64	53	44	67
-6	65	54	45	68
-5	65	54	46	68
-4	66	55	46	69
-3	66	55	47	69
-2	67	56	48	69
-1	67	57	48	70

Raw scores to standard scores Ages 8 through 9				
Recognition Raw	Verbal	Non-verbal	Semantic	Non-semantic
0	68	57	49	70
1	68	58	50	71
2	69	58	50	71
3	69	59	51	72
4	70	60	52	72
5	70	60	52	73
6	70	61	53	73
7	71	61	54	74
8	71	62	54	74
9	72	63	55	75
10	72	63	56	75
11	73	64	56	75
12	73	64	57	76
13	74	65	58	76
14	74	66	58	77
15	75	66	59	77
16	75	67	60	78
17	76	67	60	78
18	76	68	61	79
19	77	69	62	79
20	77	69	62	80
21	77	70	63	80
22	78	70	64	81
23	78	71	64	81
24	79	71	65	81
25	79	72	66	82
26	80	73	66	82
27	80	73	67	83
28	81	74	68	83
29	81	74	68	84

Recognition
Ages 8 through 9

Raw scores to standard scores Ages 8 through 9				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
30	82	75	69	84
31	82	76	70	85
32	83	76	70	85
33	83	77	71	86
34	84	77	72	86
35	84	78	72	87
36	85	79	73	87
37	85	79	74	87
38	85	80	74	88
39	86	80	75	88
40	86	81	76	89
41	87	82	76	89
42	87	82	77	90
43	88	83	78	90
44	88	83	78	91
45	89	84	79	91
46	89	85	80	92
47	90	85	80	92
48	90	86	81	93
49	91	86	82	93
50	91	87	82	93
51	92	87	83	94
52	92	88	84	94
53	93	89	84	95
54	93	89	85	95
55	93	90	86	96
56	94	90	86	96
57	94	91	87	97
58	95	92	88	97
59	95	92	88	98

Raw scores to standard scores Ages 8 through 9				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
60	96	93	89	98
61	96	93	89	99
62	97	94	90	99
63	97	95	91	99
64	98	95	91	100
65	98	96	92	100
66	99	96	93	101
67	99	97	93	101
68	100	98	94	102
69	100	98	95	102
70	100	99	95	103
71	101	99	96	103
72	101	100	97	104
73	102	101	97	104
74	102	101	98	105
75	103	102	99	105
76	103	102	99	105
77	104	103	100	106
78	104	103	101	106
79	105	104	101	107
80	105	105	102	107
81	106	105	103	108
82	106	106	103	108
83	107	106	104	109
84	107	107	105	109
85	108	108	105	110
86	108	108	106	110
87	108	109	107	111
88	109	109	107	111
89	109	110	108	111

Raw scores to standard scores Ages 8 through 9				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
90	110	111	109	112
91	110	111	109	112
92	111	112	110	113
93	111	112	111	113
94	112	113	111	114
95	112	114	112	114
96	113	114	113	115
97	113	115	113	115
98	114	115	114	116
99	114	116	115	116
100	115	117	115	116

Raw scores to standard scores Ages 10 through 11				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
1	67	64	65	69
2	68	65	66	70
3	68	66	67	71
4	69	66	67	72
5	70	67	68	73
6	71	68	69	74
7	72	69	69	74
8	72	70	70	75
9	73	71	71	76
10	74	71	71	77
11	75	72	72	78
12	76	73	73	79
13	76	74	73	80
14	77	75	74	81
15	78	76	75	82
16	79	76	75	83
17	80	77	76	84
18	80	78	77	85
19	81	79	77	86
20	82	80	78	87
21	83	81	79	88
22	83	81	79	89
23	84	82	80	90
24	85	83	81	91
25	86	84	81	92
26	87	85	82	93
27	87	86	83	94
28	88	86	83	95
29	89	87	84	96
30	90	88	85	96

Raw scores to standard scores Ages 10 through 11				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
31	91	90	86	98
32	92	91	87	99
33	93	92	87	100
34	94	92	88	101
35	95	93	89	102
36	95	94	89	103
37	96	95	90	104
38	97	96	91	105
39	98	97	91	106
40	99	97	92	107
41	99	98	93	108
42	100	99	93	109
43	101	100	94	110
44	102	101	95	111
45	103	102	95	112
46	103	102	96	113
47	104	103	97	114
48	105	104	97	115
49	106	105	98	116
50	107	106	99	117
51	107	107	99	118
52	108	107	100	118
53	109	108	101	119
54	110	109	101	120
55	111	110	102	121
56	111	111	102	122
57	112	112	103	123
58	113	112	104	124
59	114	113	104	125
60	114	114	105	126

Raw scores to standard scores Ages 10 through 11				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
61	115	115	106	127
62	116	116	106	128
63	117	117	107	129
64	118	117	108	130
65	118	118	108	131
66	119	119	109	132
67	120	120	110	133
68	121	121	110	134
69	122	122	111	135
70	122	122	112	136
71	123	123	112	137
72	124	124	113	138
73	125	125	114	139
74	126	126	114	140
75	126	127	115	140
76	127	128	116	141
77	128	128	116	142
78	129	129	117	143
79	130	130	118	144
80	130	131	118	145
81	131	132	119	146
82	132	133	120	147
83	133	133	120	148
84	134	134	121	149
85	134	135	122	150
86	135	136	122	151
87	136	137	123	152
88	137	138	124	153
89	138	138	124	154
90	138	139	125	155

Raw scores to standard scores Ages 10 through 11				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
91	139	140	126	156
92	140	141	126	157
93	141	142	127	158
94	142	143	128	159
95	142	143	128	160
96	143	144	129	161
97	144	145	130	162
98	145	146	130	162
99	145	147	131	163
100	146	148	132	164

Learning Gain

Ages 10 through 11

Raw scores to standard scores Ages 10 through 11				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
-30	49	42	52	44
-29	50	43	52	45
-28	51	44	53	46
-27	52	45	54	47
-26	52	46	55	48
-25	53	47	55	49
-24	54	48	56	50
-23	55	49	57	51
-22	56	50	58	52
-21	57	51	59	53
-20	57	53	59	54
-19	58	54	60	55
-18	59	55	61	56
-17	60	56	62	57
-16	61	57	63	58
-15	61	58	63	59
-14	62	59	64	60
-13	63	60	65	61
-12	64	61	66	62
-11	65	62	66	63
-10	65	63	67	64
-9	66	64	68	65
-8	67	65	69	66
-7	68	67	70	67
-6	69	68	70	68
-5	69	69	71	69
-4	70	70	72	70
-3	71	71	73	71
-2	72	72	73	72
-1	73	73	74	73

Raw scores to standard scores Ages 10 through 11				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	73	74	75	74
1	74	75	76	75
2	75	76	77	76
3	76	77	77	77
4	77	78	78	78
5	77	79	79	79
6	78	80	80	80
7	79	82	80	81
8	80	83	81	82
9	81	84	82	83
10	81	85	83	84
11	82	86	84	85
12	83	87	84	86
13	84	88	85	87
14	85	89	86	88
15	85	90	87	90
16	86	91	88	91
17	87	92	88	92
18	88	93	89	93
19	89	94	90	94
20	89	96	91	95
21	90	97	91	96
22	91	98	92	97
23	92	99	93	98
24	93	100	94	99
25	94	101	95	100
26	94	102	95	101
27	95	103	96	102
28	96	104	97	103
29	97	105	98	104

Learning Gain

Ages 10 through 11

Raw scores to standard scores Ages 10 through 11				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	98	106	98	105
31	98	107	99	106
32	99	108	100	107
33	100	109	101	108
34	101	111	102	109
35	102	112	102	110
36	102	113	103	111
37	103	114	104	112
38	104	115	105	113
39	105	116	106	114
40	106	117	106	115
41	106	118	107	116
42	107	119	108	117
43	108	120	109	118
44	109	121	109	119
45	110	122	110	120
46	110	123	111	121
47	111	125	112	122
48	112	126	113	123
49	113	127	113	124
50	114	128	114	125
51	114	129	115	126
52	115	130	116	127
53	116	131	116	128
54	117	132	117	129
55	118	133	118	130
56	118	134	119	131
57	119	135	120	132
58	120	136	120	133
59	121	137	121	134

Raw scores to standard scores Ages 10 through 11				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	122	139	122	135
61	122	140	123	136
62	123	141	123	137
63	124	142	124	138
64	125	143	125	140
65	126	144	126	141
66	127	145	127	142
67	127	146	127	143
68	128	147	128	144
69	129	148	129	145
70	130	149	130	146
71	131	150	131	147
72	131	151	131	148
73	132	152	132	149
74	133	154	133	150
75	134	155	134	151
76	135	156	134	152
77	135	157	135	153
78	136	158	136	154
79	137	159	137	155
80	138	160	138	156
81	139	161	138	157
82	139	162	139	158
83	140	163	140	159

Delayed recall

Ages 10 through 11

Raw scores to standard scores Ages 10 through 11				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	69	65	66	72
1	70	66	66	73
2	71	66	67	74
3	71	67	67	75
4	72	68	68	75
5	73	68	69	76
6	73	69	69	77
7	74	70	70	77
8	74	70	70	78
9	75	71	71	79
10	76	72	71	80
11	76	72	72	80
12	77	73	73	81
13	78	74	73	82
14	78	74	74	82
15	79	75	74	83
16	80	76	75	84
17	80	76	75	85
18	81	77	76	85
19	81	78	76	86
20	82	78	77	87
21	83	79	78	87
22	83	80	78	88
23	84	80	79	89
24	85	81	79	90
25	85	82	80	90
26	86	83	80	91
27	86	83	81	92
28	87	84	81	92
29	88	85	82	93

Raw scores to standard scores Ages 10 through 11				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	88	85	83	94
31	89	86	83	95
32	90	87	84	95
33	90	87	84	96
34	91	88	85	97
35	92	89	85	97
36	92	89	86	98
37	93	90	86	99
38	93	91	87	100
39	94	91	88	100
40	95	92	88	101
41	95	93	89	102
42	96	93	89	102
43	97	94	90	103
44	97	95	90	104
45	98	95	91	105
46	99	96	91	105
47	99	97	92	106
48	100	98	93	107
49	100	98	93	107
50	101	99	94	108
51	102	100	94	109
52	102	100	95	110
53	103	101	95	110
54	104	102	96	111
55	104	102	96	112
56	105	103	97	112
57	105	104	98	113
58	106	104	98	114
59	107	105	99	115

Delayed recall

Ages 10 through 11

Raw scores to standard scores Ages 10 through 11				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	107	106	99	115
61	108	106	100	116
62	109	107	100	117
63	109	108	101	117
64	110	108	101	118
65	111	109	102	119
66	111	110	103	120
67	112	110	103	120
68	112	111	104	121
69	113	112	104	122
70	114	112	105	122
71	114	113	105	123
72	115	114	106	124
73	116	115	106	125
74	116	115	107	125
75	117	116	108	126
76	118	117	108	127
77	118	117	109	127
78	119	118	109	128
79	119	119	110	129
80	120	119	110	130
81	121	120	111	130
82	121	121	111	131
83	122	121	112	132
84	123	122	113	132
85	123	123	113	133
86	124	123	114	134
87	124	124	114	135
88	125	125	115	135
89	126	125	115	136

Raw scores to standard scores Ages 10 through 11				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
90	126	126	116	137
91	127	127	116	137
92	128	127	117	138
93	128	128	118	139
94	129	129	118	140
95	130	130	119	140
96	130	130	119	141
97	131	131	120	142
98	131	132	120	142
99	132	132	121	143
100	133	133	121	144

Forgetting

Ages 10 through 11

Raw scores to standard scores Ages 10 through 11				
Forgetting Raw	Verbal	Non-verbal	Semantic	Non-semantic
-50	21	20	40	11
-49	23	22	41	13
-48	25	24	42	15
-47	26	25	44	17
-46	28	27	45	19
-45	30	29	47	21
-44	32	30	48	23
-43	34	32	49	25
-42	36	34	51	27
-41	38	36	52	29
-40	40	37	53	31
-39	42	39	55	33
-38	44	41	56	35
-37	46	43	58	37
-36	48	44	59	40
-35	50	46	60	42
-34	52	48	62	44
-33	54	49	63	46
-32	56	51	64	48
-31	58	53	66	50
-30	60	55	67	52
-29	61	56	69	54
-28	63	58	70	56
-27	65	60	71	58
-26	67	61	73	60
-25	69	63	74	62
-24	71	65	75	64
-23	73	67	77	66
-22	75	68	78	68
-21	77	70	80	70

Raw scores to standard scores Ages 10 through 11				
Forgetting Raw	Verbal	Non-verbal	Semantic	Non-semantic
-20	79	72	81	73
-19	81	74	82	75
-18	83	75	84	77
-17	85	77	85	79
-16	87	79	86	81
-15	89	80	88	83
-14	91	82	89	85
-13	93	84	91	87
-12	94	86	92	89
-11	96	87	93	91
-10	98	89	95	93
-9	100	91	96	95
-8	102	92	97	97
-7	104	94	99	99
-6	106	96	100	101
-5	108	98	102	103
-4	110	99	103	105
-3	112	101	104	108
-2	114	103	106	110
-1	116	104	107	112
0	118	106	108	114
1	120	108	110	116
2	122	110	111	118
3	124	111	113	120
4	126	113	114	122
5	128	115	115	124
6	129	117	117	126
7	131	118	118	128
8	133	120	119	130
9	135	122	121	132

Raw scores to standard scores Ages 10 through 11				
Forgetting Raw	Verbal	Non- verbal	Semantic	Non- semantic
10	137	123	122	134
11	139	125	124	136
12	141	127	125	138
13	143	129	126	141
14	145	130	128	143
15	147	132	129	145
16	149	134	130	147
17	151	135	132	149
18	153	137	133	151
19	155	139	134	153
20	157	141	136	155
21	159	142	137	157
22	161	144	139	159
23	163	146	140	161
24	164	148	141	163
25	166	149	143	165

Recognition
Ages 10 through 11

Raw scores to standard scores Ages 10 through 11				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
-30	55	45	31	59
-29	55	45	32	59
-28	56	46	32	60
-27	56	46	33	60
-26	57	47	34	61
-25	57	47	34	61
-24	58	48	35	62
-23	58	49	35	62
-22	59	49	36	62
-21	59	50	37	63
-20	59	50	37	63
-19	60	51	38	64
-18	60	51	39	64
-17	61	52	39	65
-16	61	52	40	65
-15	62	53	40	66
-14	62	54	41	66
-13	62	54	42	67
-12	63	55	42	67
-11	63	55	43	67
-10	64	56	43	68
-9	64	56	44	68
-8	65	57	45	69
-7	65	57	45	69
-6	65	58	46	70
-5	66	59	47	70
-4	66	59	47	71
-3	67	60	48	71
-2	67	60	48	72
-1	68	61	49	72

Raw scores to standard scores Ages 10 through 11				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
0	68	61	50	72
1	69	62	50	73
2	69	63	51	73
3	69	63	51	74
4	70	64	52	74
5	70	64	53	75
6	71	65	53	75
7	71	65	54	76
8	72	66	55	76
9	72	66	55	77
10	72	67	56	77
11	73	68	56	77
12	73	68	57	78
13	74	69	58	78
14	74	69	58	79
15	75	70	59	79
16	75	70	59	80
17	75	71	60	80
18	76	71	61	81
19	76	72	61	81
20	77	73	62	82
21	77	73	63	82
22	78	74	63	82
23	78	74	64	83
24	79	75	64	83
25	79	75	65	84
26	79	76	66	84
27	80	77	66	85
28	80	77	67	85
29	81	78	68	86

Recognition
Ages 10 through 11

Raw scores to standard scores Ages 10 through 11				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
30	81	78	68	86
31	82	79	69	87
32	82	79	69	87
33	82	80	70	87
34	83	80	71	88
35	83	81	71	88
36	84	82	72	89
37	84	82	72	89
38	85	83	73	90
39	85	83	74	90
40	85	84	74	91
41	86	84	75	91
42	86	85	76	92
43	87	85	76	92
44	87	86	77	92
45	88	87	77	93
46	88	87	78	93
47	89	88	79	94
48	89	88	79	94
49	89	89	80	95
50	90	89	80	95
51	90	90	81	96
52	91	91	82	96
53	91	91	82	97
54	92	92	83	97
55	92	92	84	97
56	92	93	84	98
57	93	93	85	98
58	93	94	85	99
59	94	94	86	99

Raw scores to standard scores Ages 10 through 11				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
60	94	95	87	100
61	95	96	87	100
62	95	96	88	101
63	96	97	88	101
64	96	97	89	102
65	96	98	90	102
66	97	98	90	102
67	97	99	91	103
68	98	100	92	103
69	98	100	92	104
70	99	101	93	104
71	99	101	93	105
72	99	102	94	105
73	100	102	95	106
74	100	103	95	106
75	101	103	96	107
76	101	104	97	107
77	102	105	97	107
78	102	105	98	108
79	102	106	98	108
80	103	106	99	109
81	103	107	100	109
82	104	107	100	110
83	104	108	101	110
84	105	108	101	111
85	105	109	102	111
86	106	110	103	112
87	106	110	103	112
88	106	111	104	112
89	107	111	105	113

Raw scores to standard scores Ages 10 through 11				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
90	107	112	105	113
91	108	112	106	114
92	108	113	106	114
93	109	114	107	115
94	109	114	108	115
95	109	115	108	116
96	110	115	109	116
97	110	116	109	117
98	111	116	110	117
99	111	117	111	117
100	112	117	111	118

Raw scores to standard scores Ages 12 through 13				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
1	62	56	49	58
2	63	56	50	59
3	63	57	51	60
4	64	58	52	61
5	65	59	52	62
6	65	60	53	63
7	66	61	54	64
8	67	62	55	65
9	68	62	56	67
10	68	63	56	68
11	69	64	57	69
12	70	65	58	70
13	71	66	59	71
14	71	67	60	72
15	72	68	60	73
16	73	68	61	74
17	74	69	62	75
18	74	70	63	77
19	75	71	63	78
20	76	72	64	79
21	76	73	65	80
22	77	74	66	81
23	78	74	67	82
24	79	75	67	83
25	79	76	68	84
26	80	77	69	86
27	81	78	70	87
28	82	79	70	88
29	82	80	71	89
30	83	80	72	90

Raw scores to standard scores Ages 12 through 13				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
31	83	83	80	89
32	84	84	80	90
33	85	84	81	90
34	85	85	82	91
35	86	86	82	92
36	87	86	83	92
37	87	87	84	93
38	88	88	84	94
39	89	88	85	94
40	89	89	85	95
41	90	90	86	96
42	91	90	87	96
43	92	91	87	97
44	92	92	88	98
45	93	93	88	98
46	94	93	89	99
47	94	94	90	100
48	95	95	90	100
49	96	95	91	101
50	96	96	92	102
51	97	97	92	102
52	98	97	93	103
53	98	98	93	104
54	99	99	94	104
55	100	100	95	105
56	101	100	95	106
57	101	101	96	106
58	102	102	97	107
59	103	102	97	108
60	103	103	98	108

Raw scores to standard scores Ages 12 through 13				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
61	106	107	96	125
62	107	108	97	126
63	107	109	98	127
64	108	109	99	128
65	109	110	99	129
66	109	111	100	130
67	110	112	101	131
68	111	113	102	132
69	112	114	103	134
70	112	115	103	135
71	113	115	104	136
72	114	116	105	137
73	115	117	106	138
74	115	118	106	139
75	116	119	107	140
76	117	120	108	141
77	118	121	109	143
78	118	121	110	144
79	119	122	110	145
80	120	123	111	146
81	120	124	112	147
82	121	125	113	148
83	122	126	113	149
84	123	127	114	150
85	123	127	115	151
86	124	128	116	153
87	125	129	117	154
88	126	130	117	155
89	126	131	118	156
90	127	132	119	157

Raw scores to standard scores Ages 12 through 13				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
91	123	129	115	136
92	124	130	116	137
93	125	132	118	138
94	126	133	119	139
95	127	134	120	140
96	128	136	121	141
97	129	137	122	142
98	130	138	123	143
99	130	140	124	144
100	131	141	126	145

Learning Gain

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
-30	57	31	49	46
-29	58	32	49	47
-28	59	33	50	48
-27	60	34	51	49
-26	60	35	52	50
-25	61	36	52	51
-24	62	38	53	52
-23	62	39	54	53
-22	63	40	55	54
-21	64	41	56	55
-20	65	42	56	56
-19	65	43	57	57
-18	66	44	58	58
-17	67	46	59	59
-16	67	47	60	60
-15	68	48	60	61
-14	69	49	61	62
-13	70	50	62	63
-12	70	51	63	64
-11	71	53	64	65
-10	72	54	64	66
-9	72	55	65	67
-8	73	56	66	68
-7	74	57	67	69
-6	75	58	68	70
-5	75	59	68	71
-4	76	61	69	72
-3	77	62	70	73
-2	77	63	71	74
-1	78	64	71	75

Raw scores to standard scores Ages 12 through 13				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	79	65	72	76
1	80	66	73	77
2	80	67	74	78
3	81	69	75	79
4	82	70	75	80
5	82	71	76	81
6	83	72	77	82
7	84	73	78	83
8	85	74	79	84
9	85	75	79	85
10	86	77	80	86
11	87	78	81	87
12	87	79	82	88
13	88	80	83	89
14	89	81	83	90
15	90	82	84	91
16	90	84	85	92
17	91	85	86	93
18	92	86	87	94
19	92	87	87	95
20	93	88	88	96
21	94	89	89	97
22	95	90	90	98
23	95	92	90	99
24	96	93	91	100
25	97	94	92	100
26	97	95	93	101
27	98	96	94	102
28	99	97	94	103
29	100	98	95	104

Learning Gain

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	100	100	96	105
31	101	101	97	106
32	102	102	98	107
33	102	103	98	108
34	103	104	99	109
35	104	105	100	110
36	105	107	101	111
37	105	108	102	112
38	106	109	102	113
39	107	110	103	114
40	107	111	104	115
41	108	112	105	116
42	109	113	106	117
43	110	115	106	118
44	110	116	107	119
45	111	117	108	120
46	112	118	109	121
47	112	119	110	122
48	113	120	110	123
49	114	121	111	124
50	114	123	112	125
51	115	124	113	126
52	116	125	113	127
53	117	126	114	128
54	117	127	115	129
55	118	128	116	130
56	119	129	117	131
57	119	131	117	132
58	120	132	118	133
59	121	133	119	134

Raw scores to standard scores Ages 12 through 13				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	122	134	120	135
61	122	135	121	136
62	123	136	121	137
63	124	138	122	138
64	124	139	123	139
65	125	140	124	140
66	126	141	125	141
67	127	142	125	142
68	127	143	126	143
69	128	144	127	144
70	129	146	128	145
71	129	147	129	146
72	130	148	129	147
73	131	149	130	148
74	132	150	131	149
75	132	151	132	150
76	133	152	132	151
77	134	154	133	152
78	134	155	134	153
79	135	156	135	154
80	0	0	0	0
81	0	0	0	0
82	0	0	0	0
83	0	0	0	0

Delayed recall

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	60	57	47	66
1	61	58	48	67
2	62	58	48	68
3	62	59	49	68
4	63	60	50	69
5	63	60	51	70
6	64	61	51	71
7	65	62	52	71
8	65	62	53	72
9	66	63	53	73
10	67	64	54	74
11	67	65	55	74
12	68	65	55	75
13	69	66	56	76
14	69	67	57	77
15	70	67	57	77
16	70	68	58	78
17	71	69	59	79
18	72	69	59	80
19	72	70	60	80
20	73	71	61	81
21	74	72	61	82
22	74	72	62	83
23	75	73	63	83
24	76	74	64	84
25	76	74	64	85
26	77	75	65	85
27	77	76	66	86
28	78	76	66	87
29	79	77	67	88

Raw scores to standard scores Ages 12 through 13				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	79	78	68	88
31	80	79	68	89
32	81	79	69	90
33	81	80	70	91
34	82	81	70	91
35	83	81	71	92
36	83	82	72	93
37	84	83	72	94
38	84	83	73	94
39	85	84	74	95
40	86	85	75	96
41	86	86	75	97
42	87	86	76	97
43	88	87	77	98
44	88	88	77	99
45	89	88	78	100
46	90	89	79	100
47	90	90	79	101
48	91	90	80	102
49	91	91	81	103
50	92	92	81	103
51	93	93	82	104
52	93	93	83	105
53	94	94	83	106
54	95	95	84	106
55	95	95	85	107
56	96	96	85	108
57	97	97	86	109
58	97	97	87	109
59	98	98	88	110

Delayed recall

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	98	99	88	111
61	99	100	89	112
62	100	100	90	112
63	100	101	90	113
64	101	102	91	114
65	102	102	92	115
66	102	103	92	115
67	103	104	93	116
68	103	104	94	117
69	104	105	94	117
70	105	106	95	118
71	105	107	96	119
72	106	107	96	120
73	107	108	97	120
74	107	109	98	121
75	108	109	99	122
76	109	110	99	123
77	109	111	100	123
78	110	111	101	124
79	110	112	101	125
80	111	113	102	126
81	112	114	103	126
82	112	114	103	127
83	113	115	104	128
84	114	116	105	129
85	114	116	105	129
86	115	117	106	130
87	116	118	107	131
88	116	118	107	132
89	117	119	108	132

Raw scores to standard scores Ages 12 through 13				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
90	117	120	109	133
91	118	121	109	134
92	119	121	110	135
93	119	122	111	135
94	120	123	112	136
95	121	123	112	137
96	121	124	113	138
97	122	125	114	138
98	123	125	114	139
99	123	126	115	140
100	124	127	116	141

Forgetting

Ages 12 through 13

Raw scores to standard scores Ages 12 through 13				
Forgetting Raw	Verbal	Non-verbal	Semantic	Non-semantic
-50	34	37	40	49
-49	35	39	41	50
-48	36	40	43	51
-47	38	41	44	52
-46	39	43	45	53
-45	41	44	47	54
-44	42	46	48	55
-43	44	47	49	57
-42	45	48	50	58
-41	47	50	52	59
-40	48	51	53	60
-39	49	52	54	61
-38	51	54	56	62
-37	52	55	57	63
-36	54	56	58	65
-35	55	58	59	66
-34	57	59	61	67
-33	58	61	62	68
-32	60	62	63	69
-31	61	63	65	70
-30	63	65	66	71
-29	64	66	67	73
-28	65	67	68	74
-27	67	69	70	75
-26	68	70	71	76
-25	70	71	72	77
-24	71	73	73	78
-23	73	74	75	79
-22	74	75	76	81
-21	76	77	77	82

Raw scores to standard scores Ages 12 through 13				
Forgetting Raw	Verbal	Non-verbal	Semantic	Non-semantic
-20	77	78	79	83
-19	78	80	80	84
-18	80	81	81	85
-17	81	82	82	86
-16	83	84	84	87
-15	84	85	85	89
-14	86	86	86	90
-13	87	88	88	91
-12	89	89	89	92
-11	90	90	90	93
-10	92	92	91	94
-9	93	93	93	95
-8	94	95	94	97
-7	96	96	95	98
-6	97	97	97	99
-5	99	99	98	100
-4	100	100	99	101
-3	102	101	100	102
-2	103	103	102	103
-1	105	104	103	105
0	106	105	104	106
1	107	107	106	107
2	109	108	107	108
3	110	110	108	109
4	112	111	109	110
5	113	112	111	111
6	115	114	112	113
7	116	115	113	114
8	118	116	115	115
9	119	118	116	116

Raw scores to standard scores Ages 12 through 13				
Forgetting Raw	Verbal	Non- verbal	Semantic	Non- semantic
10	121	119	117	117
11	122	120	118	118
12	123	122	120	119
13	125	123	121	121
14	126	125	122	122
15	128	126	124	123
16	129	127	125	124
17	131	129	126	125
18	132	130	127	126
19	134	131	129	127
20	135	133	130	129
21	136	134	131	130
22	138	135	132	131
23	139	137	134	132
24	141	138	135	133
25	142	139	136	134

Raw scores to standard scores Ages 12 through 13				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
-30	5	17	1	32
-29	6	18	1	33
-28	7	18	1	33
-27	8	19	1	34
-26	9	20	1	35
-25	10	21	1	35
-24	10	21	1	36
-23	11	22	1	37
-22	12	23	1	37
-21	13	24	1	38
-20	14	24	1	38
-19	15	25	1	39
-18	15	26	1	40
-17	16	26	1	40
-16	17	27	1	41
-15	18	28	1	42
-14	19	29	1	42
-13	19	29	1	43
-12	20	30	2	44
-11	21	31	2	44
-10	22	32	3	45
-9	23	32	4	45
-8	24	33	5	46
-7	24	34	6	47
-6	25	35	7	47
-5	26	35	8	48
-4	27	36	9	49
-3	28	37	10	49
-2	29	38	11	50
-1	29	38	12	51

Raw scores to standard scores Ages 12 through 13				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
0	30	39	13	51
1	31	40	14	52
2	32	40	15	52
3	33	41	16	53
4	34	42	17	54
5	34	43	18	54
6	35	43	19	55
7	36	44	20	56
8	37	45	21	56
9	38	46	22	57
10	39	46	23	58
11	39	47	24	58
12	40	48	25	59
13	41	49	26	59
14	42	49	27	60
15	43	50	28	61
16	43	51	29	61
17	44	52	30	62
18	45	52	30	63
19	46	53	31	63
20	47	54	32	64
21	48	55	33	64
22	48	55	34	65
23	49	56	35	66
24	50	57	36	66
25	51	57	37	67
26	52	58	38	68
27	53	59	39	68
28	53	60	40	69
29	54	60	41	70

Recognition
Ages 12 through 13

Raw scores to standard scores Ages 12 through 13				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
30	55	61	42	70
31	56	62	43	71
32	57	63	44	71
33	58	63	45	72
34	58	64	46	73
35	59	65	47	73
36	60	66	48	74
37	61	66	49	75
38	62	67	50	75
39	63	68	51	76
40	63	69	52	77
41	64	69	53	77
42	65	70	54	78
43	66	71	55	78
44	67	71	56	79
45	67	72	57	80
46	68	73	58	80
47	69	74	58	81
48	70	74	59	82
49	71	75	60	82
50	72	76	61	83
51	72	77	62	84
52	73	77	63	84
53	74	78	64	85
54	75	79	65	85
55	76	80	66	86
56	77	80	67	87
57	77	81	68	87
58	78	82	69	88
59	79	83	70	89

Raw scores to standard scores Ages 12 through 13				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
60	80	83	71	89
61	81	84	72	90
62	82	85	73	90
63	82	85	74	91
64	83	86	75	92
65	84	87	76	92
66	85	88	77	93
67	86	88	78	94
68	86	89	79	94
69	87	90	80	95
70	88	91	81	96
71	89	91	82	96
72	90	92	83	97
73	91	93	84	97
74	91	94	85	98
75	92	94	86	99
76	93	95	86	99
77	94	96	87	100
78	95	97	88	101
79	96	97	89	101
80	96	98	90	102
81	97	99	91	103
82	98	100	92	103
83	99	100	93	104
84	100	101	94	104
85	101	102	95	105
86	101	102	96	106
87	102	103	97	106
88	103	104	98	107
89	104	105	99	108

Raw scores to standard scores Ages 12 through 13				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
90	105	105	100	108
91	106	106	101	109
92	106	107	102	110
93	107	108	103	110
94	108	108	104	111
95	109	109	105	111
96	110	110	106	112
97	110	111	107	113
98	111	111	108	113
99	112	112	109	114
100	113	113	110	115

Raw scores to standard scores Ages 14 through 15				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
1	62	62	61	69
2	63	63	62	70
3	64	63	62	71
4	64	64	63	71
5	65	65	64	72
6	66	65	64	73
7	66	66	65	73
8	67	67	66	74
9	68	68	66	75
10	69	68	67	75
11	69	69	67	76
12	70	70	68	77
13	71	70	69	77
14	71	71	69	78
15	72	72	70	79
16	73	72	70	79
17	73	73	71	80
18	74	74	72	81
19	75	74	72	81
20	75	75	73	82
21	76	76	74	83
22	77	77	74	83
23	78	77	75	84
24	78	78	75	85
25	79	79	76	85
26	80	79	77	86
27	80	80	77	87
28	81	81	78	87
29	82	81	79	88
30	82	82	79	88

Raw scores to standard scores Ages 14 through 15				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
31	83	83	80	89
32	84	84	80	90
33	85	84	81	90
34	85	85	82	91
35	86	86	82	92
36	87	86	83	92
37	87	87	84	93
38	88	88	84	94
39	89	88	85	94
40	89	89	85	95
41	90	90	86	96
42	91	90	87	96
43	92	91	87	97
44	92	92	88	98
45	93	93	88	98
46	94	93	89	99
47	94	94	90	100
48	95	95	90	100
49	96	95	91	101
50	96	96	92	102
51	97	97	92	102
52	98	97	93	103
53	98	98	93	104
54	99	99	94	104
55	100	100	95	105
56	101	100	95	106
57	101	101	96	106
58	102	102	97	107
59	103	102	97	108
60	103	103	98	108

Raw scores to standard scores Ages 14 through 15				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
61	104	104	98	109
62	105	104	99	110
63	105	105	100	110
64	106	106	100	111
65	107	106	101	112
66	108	107	102	112
67	108	108	102	113
68	109	109	103	114
69	110	109	103	114
70	110	110	104	115
71	111	111	105	116
72	112	111	105	116
73	112	112	106	117
74	113	113	107	118
75	114	113	107	118
76	115	114	108	119
77	115	115	108	120
78	116	116	109	120
79	117	116	110	121
80	117	117	110	122
81	118	118	111	122
82	119	118	111	123
83	119	119	112	124
84	120	120	113	124
85	121	120	113	125
86	121	121	114	126
87	122	122	115	126
88	123	122	115	127
89	124	123	116	128
90	124	124	116	128

Raw scores to standard scores Ages 14 through 15				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
91	125	125	117	129
92	126	125	118	130
93	126	126	118	130
94	127	127	119	131
95	128	127	120	132
96	128	128	120	132
97	129	129	121	133
98	130	129	121	134
99	131	130	122	134
100	131	131	123	135

Learning Gain

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
-30	41	25	37	41
-29	42	26	38	43
-28	43	27	39	44
-27	44	28	40	45
-26	45	30	41	46
-25	46	31	42	47
-24	46	32	43	48
-23	47	34	44	49
-22	48	35	45	50
-21	49	36	46	51
-20	50	37	47	52
-19	51	39	48	53
-18	52	40	49	54
-17	53	41	50	55
-16	54	43	51	56
-15	55	44	52	57
-14	56	45	53	58
-13	56	46	54	59
-12	57	48	54	60
-11	58	49	55	61
-10	59	50	56	62
-9	60	51	57	63
-8	61	53	58	64
-7	62	54	59	65
-6	63	55	60	66
-5	64	57	61	67
-4	65	58	62	68
-3	66	59	63	69
-2	67	60	64	70
-1	67	62	65	71

Raw scores to standard scores Ages 14 through 15				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	68	63	66	72
1	69	64	67	73
2	70	66	68	74
3	71	67	69	75
4	72	68	70	76
5	73	69	71	77
6	74	71	72	78
7	75	72	73	79
8	76	73	74	80
9	77	75	74	81
10	77	76	75	82
11	78	77	76	83
12	79	78	77	84
13	80	80	78	85
14	81	81	79	86
15	82	82	80	87
16	83	83	81	88
17	84	85	82	89
18	85	86	83	90
19	86	87	84	91
20	87	89	85	92
21	87	90	86	93
22	88	91	87	94
23	89	92	88	95
24	90	94	89	96
25	91	95	90	97
26	92	96	91	98
27	93	98	92	99
28	94	99	93	100
29	95	100	94	101

Learning Gain

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	96	101	94	102
31	97	103	95	103
32	98	104	96	104
33	98	105	97	105
34	99	106	98	106
35	100	108	99	107
36	101	109	100	108
37	102	110	101	109
38	103	112	102	110
39	104	113	103	111
40	105	114	104	112
41	106	115	105	113
42	107	117	106	114
43	108	118	107	115
44	108	119	108	116
45	109	121	109	117
46	110	122	110	118
47	111	123	111	119
48	112	124	112	120
49	113	126	113	121
50	114	127	114	122
51	115	128	114	123
52	116	129	115	124
53	117	131	116	125
54	118	132	117	126
55	118	133	118	127
56	119	135	119	128
57	120	136	120	129
58	121	137	121	130
59	122	138	122	131

Raw scores to standard scores Ages 14 through 15				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	123	140	123	132
61	124	141	124	133
62	125	142	125	134
63	126	144	126	135
64	127	145	127	136
65	128	146	128	137
66	128	147	129	139
67	129	149	130	140
68	130	150	131	141
69	131	151	132	142
70	132	152	133	143

Delayed recall

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	61	56	55	56
1	62	57	56	57
2	62	57	56	57
3	63	58	57	58
4	64	59	57	59
5	64	60	58	60
6	65	60	59	60
7	65	61	59	61
8	66	62	60	62
9	67	62	60	62
10	67	63	61	63
11	68	64	62	64
12	68	64	62	64
13	69	65	63	65
14	69	66	63	66
15	70	66	64	66
16	71	67	65	67
17	71	68	65	68
18	72	68	66	68
19	72	69	66	69
20	73	70	67	70
21	74	70	67	70
22	74	71	68	71
23	75	72	69	72
24	75	72	69	72
25	76	73	70	73
26	77	74	70	74
27	77	74	71	74
28	78	75	72	75
29	78	76	72	76

Raw scores to standard scores Ages 14 through 15				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	79	76	73	76
31	79	77	73	77
32	80	78	74	78
33	81	78	75	78
34	81	79	75	79
35	82	80	76	80
36	82	81	76	81
37	83	81	77	81
38	84	82	78	82
39	84	83	78	83
40	85	83	79	83
41	85	84	79	84
42	86	85	80	85
43	86	85	81	85
44	87	86	81	86
45	88	87	82	87
46	88	87	82	87
47	89	88	83	88
48	89	89	84	89
49	90	89	84	89
50	91	90	85	90
51	91	91	85	91
52	92	91	86	91
53	92	92	87	92
54	93	93	87	93
55	94	93	88	93
56	94	94	88	94
57	95	95	89	95
58	95	95	90	95
59	96	96	90	96

Delayed recall

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	96	97	91	97
61	97	97	91	97
62	98	98	92	98
63	98	99	93	99
64	99	100	93	100
65	99	100	94	100
66	100	101	94	101
67	101	102	95	102
68	101	102	96	102
69	102	103	96	103
70	102	104	97	104
71	103	104	97	104
72	103	105	98	105
73	104	106	98	106
74	105	106	99	106
75	105	107	100	107
76	106	108	100	108
77	106	108	101	108
78	107	109	101	109
79	108	110	102	110
80	108	110	103	110
81	109	111	103	111
82	109	112	104	112
83	110	112	104	112
84	111	113	105	113
85	111	114	106	114
86	112	114	106	114
87	112	115	107	115
88	113	116	107	116
89	113	116	108	116

Raw scores to standard scores Ages 14 through 15				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
90	114	117	109	117
91	115	118	109	118
92	115	118	110	118
93	116	119	110	119
94	116	120	111	120
95	117	121	112	121
96	118	121	112	121
97	118	122	113	122
98	119	123	113	123
99	119	123	114	123
100	120	124	115	124

Forgetting

Ages 14 through 15

Raw scores to standard scores Ages 14 through 15				
Forgetting Raw	Verbal	Non-verbal	Semantic	Non-semantic
-50	33	14	13	14
-49	35	16	15	16
-48	36	18	16	18
-47	38	20	18	20
-46	39	21	20	21
-45	41	23	22	23
-44	42	25	24	25
-43	44	27	26	27
-42	45	29	28	29
-41	47	30	30	30
-40	48	32	32	32
-39	50	34	34	34
-38	51	36	36	36
-37	52	38	37	38
-36	54	40	39	40
-35	55	41	41	41
-34	57	43	43	43
-33	58	45	45	45
-32	60	47	47	47
-31	61	49	49	49
-30	63	51	51	51
-29	64	52	53	52
-28	66	54	55	54
-27	67	56	57	56
-26	69	58	58	58
-25	70	60	60	60
-24	71	61	62	61
-23	73	63	64	63
-22	74	65	66	65
-21	76	67	68	67

Raw scores to standard scores Ages 14 through 15				
Forgetting Raw	Verbal	Non-verbal	Semantic	Non-semantic
-20	77	69	70	69
-19	79	71	72	71
-18	80	72	74	72
-17	82	74	76	74
-16	83	76	78	76
-15	85	78	79	78
-14	86	80	81	80
-13	88	82	83	82
-12	89	83	85	83
-11	90	85	87	85
-10	92	87	89	87
-9	93	89	91	89
-8	95	91	93	91
-7	96	93	95	93
-6	98	94	97	94
-5	99	96	98	96
-4	101	98	100	98
-3	102	100	102	100
-2	104	102	104	102
-1	105	103	106	103
0	107	105	108	105
1	108	107	110	107
2	109	109	112	109
3	111	111	114	111
4	112	113	116	113
5	114	114	118	114
6	115	116	119	116
7	117	118	121	118
8	118	120	123	120
9	120	122	125	122

Raw scores to standard scores Ages 14 through 15				
Forgetting Raw	Verbal	Non- verbal	Semantic	Non- semantic
10	121	124	127	124
11	123	125	129	125
12	124	127	131	127
13	126	129	133	129
14	127	131	135	131
15	128	133	137	133
16	130	134	139	134
17	131	136	140	136
18	133	138	142	138
19	134	140	144	140

Recognition
Ages 14 through 15

Raw scores to standard scores Ages 14 through 15				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
-30	41	20	21	45
-29	41	21	22	45
-28	42	22	23	46
-27	42	22	23	46
-26	43	23	24	47
-25	43	24	25	47
-24	44	24	25	48
-23	45	25	26	48
-22	45	26	27	49
-21	46	26	27	49
-20	46	27	28	50
-19	47	28	28	50
-18	47	28	29	51
-17	48	29	30	51
-16	48	30	30	52
-15	49	30	31	52
-14	49	31	32	53
-13	50	32	32	53
-12	50	32	33	54
-11	51	33	34	54
-10	51	34	34	55
-9	52	35	35	55
-8	53	35	36	56
-7	53	36	36	56
-6	54	37	37	57
-5	54	37	38	57
-4	55	38	38	58
-3	55	39	39	58
-2	56	39	40	59
-1	56	40	40	59

Raw scores to standard scores Ages 14 through 15				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
0	57	41	41	60
1	57	41	42	60
2	58	42	42	61
3	58	43	43	61
4	59	43	43	62
5	60	44	44	63
6	60	45	45	63
7	61	45	45	64
8	61	46	46	64
9	62	47	47	65
10	62	47	47	65
11	63	48	48	66
12	63	49	49	66
13	64	49	49	67
14	64	50	50	67
15	65	51	51	68
16	65	51	51	68
17	66	52	52	69
18	66	53	53	69
19	67	54	53	70
20	68	54	54	70
21	68	55	55	71
22	69	56	55	71
23	69	56	56	72
24	70	57	57	72
25	70	58	57	73
26	71	58	58	73
27	71	59	59	74
28	72	60	59	74
29	72	60	60	75

Recognition
Ages 14 through 15

Raw scores to standard scores Ages 14 through 15				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
30	73	61	60	75
31	73	62	61	76
32	74	62	62	76
33	75	63	62	77
34	75	64	63	77
35	76	64	64	78
36	76	65	64	78
37	77	66	65	79
38	77	66	66	79
39	78	67	66	80
40	78	68	67	81
41	79	68	68	81
42	79	69	68	82
43	80	70	69	82
44	80	71	70	83
45	81	71	70	83
46	81	72	71	84
47	82	73	72	84
48	83	73	72	85
49	83	74	73	85
50	84	75	74	86
51	84	75	74	86
52	85	76	75	87
53	85	77	75	87
54	86	77	76	88
55	86	78	77	88
56	87	79	77	89
57	87	79	78	89
58	88	80	79	90
59	88	81	79	90

Raw scores to standard scores Ages 14 through 15				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
60	89	81	80	91
61	90	82	81	91
62	90	83	81	92
63	91	83	82	92
64	91	84	83	93
65	92	85	83	93
66	92	85	84	94
67	93	86	85	94
68	93	87	85	95
69	94	88	86	95
70	94	88	87	96
71	95	89	87	96
72	95	90	88	97
73	96	90	89	97
74	97	91	89	98
75	97	92	90	99
76	98	92	91	99
77	98	93	91	100
78	99	94	92	100
79	99	94	92	101
80	100	95	93	101
81	100	96	94	102
82	101	96	94	102
83	101	97	95	103
84	102	98	96	103
85	102	98	96	104
86	103	99	97	104
87	103	100	98	105
88	104	100	98	105
89	105	101	99	106

Raw scores to standard scores Ages 14 through 15				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
90	105	105	105	105
91	105	105	105	105
92	105	105	105	105
93	105	105	105	105
94	105	105	105	105
95	105	105	105	105
96	105	105	105	105
97	105	105	105	105
98	105	105	105	105
99	105	105	105	105
100	105	105	105	105

Raw scores to standard scores Ages 16 through 18				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
1	46	9	12	46
2	47	11	13	47
3	48	12	14	48
4	49	13	15	49
5	49	15	16	50
6	50	16	17	51
7	51	17	19	52
8	52	19	20	53
9	53	20	21	54
10	54	21	22	55
11	55	23	23	56
12	55	24	24	57
13	56	25	25	58
14	57	27	27	59
15	58	28	28	60
16	59	29	29	61
17	60	31	30	62
18	61	32	31	63
19	61	33	32	64
20	62	35	34	65
21	63	36	35	66
22	64	37	36	67
23	65	39	37	68
24	66	40	38	69
25	67	41	39	70
26	67	43	40	71
27	68	44	42	72
28	69	45	43	73
29	70	47	44	74
30	71	48	45	75

Raw scores to standard scores Ages 16 through 18				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
31	72	49	46	76
32	73	51	47	77
33	74	52	48	78
34	74	53	50	79
35	75	55	51	80
36	76	56	52	81
37	77	57	53	82
38	78	59	54	83
39	79	60	55	84
40	80	61	57	85
41	80	63	58	86
42	81	64	59	87
43	82	65	60	88
44	83	67	61	89
45	84	68	62	90
46	85	69	63	91
47	86	70	65	92
48	86	72	66	93
49	87	73	67	94
50	88	74	68	95
51	89	76	69	96
52	90	77	70	97
53	91	78	72	98
54	92	80	73	99
55	92	81	74	100
56	93	82	75	101
57	94	84	76	102
58	95	85	77	103
59	96	86	78	104
60	97	88	80	105

Raw scores to standard scores Ages 16 through 18				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
61	98	89	81	106
62	99	90	82	107
63	99	92	83	108
64	100	93	84	109
65	101	94	85	110
66	102	96	86	111
67	103	97	88	112
68	104	98	89	113
69	105	100	90	114
70	105	101	91	115
71	106	102	92	116
72	107	104	93	117
73	108	105	95	118
74	109	106	96	119
75	110	108	97	120
76	111	109	98	121
77	111	110	99	122
78	112	112	100	123
79	113	113	101	124
80	114	114	103	125
81	115	116	104	126
82	116	117	105	127
83	117	118	106	128
84	117	120	107	129
85	118	121	108	130
86	119	122	109	131
87	120	124	111	132
88	121	125	112	133
89	122	126	113	134
90	123	128	114	135

Raw scores to standard scores Ages 16 through 18				
Learning Raw	Verbal	Non-verbal	Semantic	Non-semantic
91	123	129	115	136
92	124	130	116	137
93	125	132	118	138
94	126	133	119	139
95	127	134	120	140
96	128	136	121	141
97	129	137	122	142
98	130	138	123	143
99	130	140	124	144
100	131	141	126	145

Raw scores to standard scores Ages 16 through 18				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
-30	38	8	30	22
-29	39	10	31	24
-28	40	11	32	25
-27	41	13	33	26
-26	42	14	35	27
-25	43	16	36	28
-24	44	17	37	29
-23	45	19	38	31
-22	46	20	39	32
-21	47	21	40	33
-20	48	23	41	34
-19	49	24	42	35
-18	49	26	43	36
-17	50	27	44	38
-16	51	29	45	39
-15	52	30	47	40
-14	53	32	48	41
-13	54	33	49	42
-12	55	34	50	44
-11	56	36	51	45
-10	57	37	52	46
-9	58	39	53	47
-8	59	40	54	48
-7	60	42	55	49
-6	61	43	56	51
-5	61	44	57	52
-4	62	46	58	53
-3	63	47	60	54
-2	64	49	61	55
-1	65	50	62	56

Raw scores to standard scores Ages 16 through 18				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	66	52	63	58
1	67	53	64	59
2	68	55	65	60
3	69	56	66	61
4	70	57	67	62
5	71	59	68	63
6	72	60	69	65
7	73	62	70	66
8	73	63	72	67
9	74	65	73	68
10	75	66	74	69
11	76	67	75	70
12	77	69	76	72
13	78	70	77	73
14	79	72	78	74
15	80	73	79	75
16	81	75	80	76
17	82	76	81	77
18	83	78	82	79
19	84	79	83	80
20	85	80	85	81
21	85	82	86	82
22	86	83	87	83
23	87	85	88	84
24	88	86	89	86
25	89	88	90	87
26	90	89	91	88
27	91	90	92	89
28	92	92	93	90
29	93	93	94	91

Raw scores to standard scores Ages 16 through 18				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	94	95	95	93
31	95	96	97	94
32	96	98	98	95
33	97	99	99	96
34	97	101	100	97
35	98	102	101	99
36	99	103	102	100
37	100	105	103	101
38	101	106	104	102
39	102	108	105	103
40	103	109	106	104
41	104	111	107	106
42	105	112	108	107
43	106	114	110	108
44	107	115	111	109
45	108	116	112	110
46	109	118	113	111
47	109	119	114	113
48	110	121	115	114
49	111	122	116	115
50	112	124	117	116
51	113	125	118	117
52	114	126	119	118
53	115	128	120	120
54	116	129	122	121
55	117	131	123	122
56	118	132	124	123
57	119	134	125	124
58	120	135	126	125
59	121	137	127	127

Raw scores to standard scores Ages 16 through 18				
Learning gain	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	121	138	128	128
61	122	139	129	129
62	123	141	130	130
63	124	142	131	131
64	125	144	132	132
65	126	145	133	134
66	127	147	135	135
67	128	148	136	136
68	129	149	137	137
69	130	151	138	138
70	131	152	139	139
71	132	154	140	141
72	133	155	141	142
73	133	157	142	143
74	134	158	143	144
75	135	160	144	145
76	136	161	145	146
77	137	162	147	148
78	138	164	148	149
79	139	165	149	150
80	140	167	150	151
81	141	168	151	152
82	142	170	152	153
83	143	171	153	155

Delayed recall

Ages 16 through 18

Raw scores to standard scores Ages 16 through 18				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
0	45	11	1	11
1	46	12	2	12
2	47	13	3	13
3	47	14	4	14
4	48	15	5	15
5	49	16	6	16
6	50	17	7	17
7	50	18	9	18
8	51	19	10	19
9	52	20	11	20
10	53	21	12	21
11	53	22	13	22
12	54	24	14	24
13	55	25	15	25
14	55	26	16	26
15	56	27	17	27
16	57	28	19	28
17	58	29	20	29
18	58	30	21	30
19	59	31	22	31
20	60	32	23	32
21	61	33	24	33
22	61	34	25	34
23	62	35	26	35
24	63	37	27	37
25	64	38	28	38
26	64	39	30	39
27	65	40	31	40
28	66	41	32	41
29	67	42	33	42

Raw scores to standard scores Ages 16 through 18				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
30	67	43	34	43
31	68	44	35	44
32	69	45	36	45
33	69	46	37	46
34	70	47	38	47
35	71	48	40	48
36	72	50	41	50
37	72	51	42	51
38	73	52	43	52
39	74	53	44	53
40	75	54	45	54
41	75	55	46	55
42	76	56	47	56
43	77	57	48	57
44	78	58	49	58
45	78	59	51	59
46	79	60	52	60
47	80	62	53	62
48	81	63	54	63
49	81	64	55	64
50	82	65	56	65
51	83	66	57	66
52	83	67	58	67
53	84	68	59	68
54	85	69	60	69
55	86	70	62	70
56	86	71	63	71
57	87	72	64	72
58	88	73	65	73
59	89	75	66	75

Delayed recall

Ages 16 through 18

Raw scores to standard scores Ages 16 through 18				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
60	89	76	67	76
61	90	77	68	77
62	91	78	69	78
63	92	79	70	79
64	92	80	72	80
65	93	81	73	81
66	94	82	74	82
67	95	83	75	83
68	95	84	76	84
69	96	85	77	85
70	97	86	78	86
71	97	88	79	88
72	98	89	80	89
73	99	90	81	90
74	100	91	83	91
75	100	92	84	92
76	101	93	85	93
77	102	94	86	94
78	103	95	87	95
79	103	96	88	96
80	104	97	89	97
81	105	98	90	98
82	106	100	91	100
83	106	101	93	101
84	107	102	94	102
85	108	103	95	103
86	109	104	96	104
87	109	105	97	105
88	110	106	98	106
89	111	107	99	107

Raw scores to standard scores Ages 16 through 18				
Delayed recall	Verbal	Non-verbal	Semantic	Non-semantic
Raw				
90	111	108	100	108
91	112	109	101	109
92	113	110	102	110
93	114	111	104	111
94	114	113	105	113
95	115	114	106	114
96	116	115	107	115
97	117	116	108	116
98	117	117	109	117
99	118	118	110	118
100	119	119	111	119

Forgetting

Ages 16 through 18

Raw scores to standard scores Ages 16 through 18				
Forgetting Raw	Verbal	Non-verbal	Semantic	Non-semantic
-50	13	5	1	5
-49	15	7	1	7
-48	17	9	1	9
-47	19	11	1	11
-46	21	13	1	13
-45	23	15	1	15
-44	25	17	1	17
-43	27	19	1	19
-42	29	21	1	21
-41	31	23	1	23
-40	33	25	1	25
-39	35	27	1	27
-38	37	29	2	29
-37	38	31	4	31
-36	40	33	7	33
-35	42	35	10	35
-34	44	37	13	37
-33	46	38	15	38
-32	48	40	18	40
-31	50	42	21	42
-30	52	44	24	44
-29	54	46	26	46
-28	56	48	29	48
-27	58	50	32	50
-26	60	52	35	52
-25	62	54	37	54
-24	64	56	40	56
-23	65	58	43	58
-22	67	60	46	60
-21	69	62	48	62

Raw scores to standard scores Ages 16 through 18				
Forgetting Raw	Verbal	Non-verbal	Semantic	Non-semantic
-20	71	64	51	64
-19	73	66	54	66
-18	75	68	56	68
-17	77	70	59	70
-16	79	72	62	72
-15	81	74	65	74
-14	83	76	67	76
-13	85	78	70	78
-12	87	80	73	80
-11	89	81	76	81
-10	91	83	78	83
-9	93	85	81	85
-8	94	87	84	87
-7	96	89	87	89
-6	98	91	89	91
-5	100	93	92	93
-4	102	95	95	95
-3	104	97	98	97
-2	106	99	100	99
-1	108	101	103	101
0	110	103	106	103
1	112	105	108	105
2	114	107	111	107
3	116	109	114	109
4	118	111	117	111
5	120	113	119	113
6	121	115	122	115
7	123	117	125	117
8	125	119	128	119
9	127	121	130	121

Raw scores to standard scores Ages 16 through 18				
Forgetting Raw	Verbal	Non- verbal	Semantic	Non- semantic
10	129	122	133	122
11	131	124	136	124
12	133	126	139	126
13	135	128	141	128
14	137	130	144	130
15	139	132	147	132
16	141	134	150	134
17	143	136	152	136
18	145	138	155	138
19	147	140	158	140

Recognition
Ages 16 through 18

Raw scores to standard scores Ages 16 through 18				
Recognition Raw	Verbal	Non-verbal	Semantic	Non-semantic
-30	1	1	1	2
-29	1	2	1	3
-28	1	2	1	4
-27	1	3	1	5
-26	1	4	1	6
-25	1	5	1	6
-24	1	6	1	7
-23	1	6	1	8
-22	1	7	1	9
-21	1	8	1	10
-20	1	9	1	11
-19	1	10	1	11
-18	2	11	1	12
-17	3	11	2	13
-16	4	12	3	14
-15	5	13	4	15
-14	6	14	5	16
-13	7	15	5	16
-12	7	15	6	17
-11	8	16	7	18
-10	9	17	8	19
-9	10	18	9	20
-8	11	19	10	21
-7	12	20	11	21
-6	13	20	12	22
-5	14	21	12	23
-4	15	22	13	24
-3	16	23	14	25
-2	17	24	15	26
-1	17	24	16	26

Raw scores to standard scores Ages 16 through 18				
Recognition Raw	Verbal	Non-verbal	Semantic	Non-semantic
0	18	25	17	27
1	19	26	18	28
2	20	27	19	29
3	21	28	19	30
4	22	28	20	31
5	23	29	21	31
6	24	30	22	32
7	25	31	23	33
8	26	32	24	34
9	26	33	25	35
10	27	33	25	36
11	28	34	26	36
12	29	35	27	37
13	30	36	28	38
14	31	37	29	39
15	32	37	30	40
16	33	38	31	41
17	34	39	32	41
18	35	40	32	42
19	35	41	33	43
20	36	42	34	44
21	37	42	35	45
22	38	43	36	46
23	39	44	37	46
24	40	45	38	47
25	41	46	39	48
26	42	46	39	49
27	43	47	40	50
28	44	48	41	51
29	45	49	42	52

Recognition
Ages 16 through 18

Raw scores to standard scores Ages 16 through 18				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
30	45	50	43	52
31	46	51	44	53
32	47	51	45	54
33	48	52	46	55
34	49	53	46	56
35	50	54	47	57
36	51	55	48	57
37	52	55	49	58
38	53	56	50	59
39	54	57	51	60
40	54	58	52	61
41	55	59	52	62
42	56	59	53	62
43	57	60	54	63
44	58	61	55	64
45	59	62	56	65
46	60	63	57	66
47	61	64	58	67
48	62	64	59	67
49	63	65	59	68
50	63	66	60	69
51	64	67	61	70
52	65	68	62	71
53	66	68	63	72
54	67	69	64	72
55	68	70	65	73
56	69	71	66	74
57	70	72	66	75
58	71	73	67	76
59	72	73	68	77

Raw scores to standard scores Ages 16 through 18				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
60	72	74	69	77
61	73	75	70	78
62	74	76	71	79
63	75	77	72	80
64	76	77	72	81
65	77	78	73	82
66	78	79	74	82
67	79	80	75	83
68	80	81	76	84
69	81	82	77	85
70	82	82	78	86
71	82	83	79	87
72	83	84	79	87
73	84	85	80	88
74	85	86	81	89
75	86	86	82	90
76	87	87	83	91
77	88	88	84	92
78	89	89	85	92
79	90	90	86	93
80	91	91	86	94
81	91	91	87	95
82	92	92	88	96
83	93	93	89	97
84	94	94	90	98
85	95	95	91	98
86	96	95	92	99
87	97	96	93	100
88	98	97	93	101
89	99	98	94	102

Raw scores to standard scores Ages 16 through 18				
Recognition Raw	Verbal	Non- verbal	Semantic	Non- semantic
90	100	99	95	103
91	100	99	96	103
92	101	100	97	104
93	102	101	98	105
94	103	102	99	106
95	104	103	99	107
96	105	104	100	108
97	106	104	101	108
98	107	105	102	109
99	108	106	103	110
100	109	107	104	111